

PROJECT SYNOPSIS

Accident Prone Road Prediction System Using Machine Learning

1. Title

Accident Prone Road Prediction System Using Machine Learning

Team Details:

Student Name	College Name	Branch	Mobile No	Mail ID	Role
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Program Track and Domain:

Program Track

Domain

Science-HUB

Machine Learning

Guide Details:

Project Guide: Thribhuvan Reddy M

Designation: Subject Matter Expert (SME)

2. Problem Statement

Road accidents are influenced by multiple factors such as road type, weather, visibility, traffic density, temperature, number of lanes, traffic signals, and peak-hour conditions. Identifying risky road conditions in advance is difficult when these factors are considered separately. The project addresses this problem by using historical Indian road-accident data to identify patterns associated with different accident-risk levels and provide a risk classification for given road conditions.

3. Objectives

- To analyze road and environmental factors associated with accident risk.
- To preprocess relevant categorical and numerical road-condition data.
- To develop a machine learning model for classifying accident risk as Low, Medium, or High.
- To evaluate the model using a train-test split and accuracy measurement.
- To provide a simple web interface for entering road conditions and viewing the predicted risk and safety

recommendation.

- To deploy the Flask-based prototype so that the prediction system can be accessed through the web.

4. Methodology / Implementation Plan

The system follows a machine-learning workflow consisting of data loading, preprocessing, feature selection, categorical encoding, model training, evaluation, model saving, web integration, and deployment.

- Dataset: An Indian road-accident dataset containing road, traffic, weather, and accident-related attributes.
- Preprocessing: Select relevant input features and convert categorical values into numerical form using LabelEncoder.
- Risk classification: Convert the existing risk score into Low, Medium, and High risk categories.
- Model: Random Forest Classifier with 100 decision trees.
- Evaluation: Split the data into 80% training and 20% testing data and calculate classification accuracy.
- Model persistence: Save the trained model and encoders as .pkl files using Joblib.
- Application: Use Flask to receive user inputs, apply the saved encoders, make predictions, and display the result through HTML templates.
- Deployment: Deploy the Flask web application using Render and connect it to the GitHub repository.

5. Resources and Timeline

A. Hardware & Software Requirements

- Laptop/PC with internet connectivity.
- Operating System: Windows.
- Programming Language: Python.
- Development Environment: Visual Studio Code.
- Libraries/Tools: Flask, Pandas, NumPy, Scikit-learn, Joblib, Matplotlib and Seaborn.
- Version Control: Git and GitHub.
- Deployment Platform: Render.

B. Dataset

Dataset availability: Available. The project uses an Indian road-accident dataset with 20,000 records and 24 columns. The dataset includes attributes such as city, state, latitude, longitude, road type, traffic signal, weather, visibility, temperature, traffic density, lanes, peak-hour status, accident severity, casualties, and risk score. The project dataset is maintained in the repository as dataset/indian_roads_dataset.csv. No external source URL was provided in the supplied project materials.

C. Project Timeline

Phase	Task	Tentative Completion
1	Dataset collection and understanding	Completed
2	Data preprocessing and feature selection	Completed
3	Model training and evaluation	Completed
4	Flask web application development	Completed
5	Testing with Low/Medium/High risk inputs	Completed

6

GitHub repository setup and
deployment

Completed

7

Final documentation and presentation

As per academic schedule

6. Expected Outcomes / Conclusion

The project is expected to provide a functional machine learning-based accident-risk classification system. The trained Random Forest model classifies road conditions into Low, Medium, or High risk categories. The web application provides a user-friendly prediction form and displays the predicted risk level, a risk percentage indicator, and a safety recommendation. The project also provides dashboard visualizations and a deployed prototype that demonstrates the complete workflow from user input to prediction.

7. References

- Flask Documentation — <https://flask.palletsprojects.com/>
- Pandas Documentation — <https://pandas.pydata.org/docs/>
- Scikit-learn Documentation — <https://scikit-learn.org/stable/>
- Joblib Documentation — <https://joblib.readthedocs.io/>
- Matplotlib Documentation — <https://matplotlib.org/stable/>
- Seaborn Documentation — <https://seaborn.pydata.org/>
- Git Documentation — <https://git-scm.com/doc>